Can a QPSK link reliably carry image traffic over fading channels?

Communication Monte Carlo experiment · theory validation · Rican fading · coding gain

BITS

source bits

QPSK

2 bits / symbol

CHANNEL

h·s + n

RX

BER / image

MAIN QUESTION

First prove the link simulator is trustworthy

COMMUNICATION

Channel effects + coding gain

APPLICATION

CV only allocates source bits

Open the presentation with this sentence

Explain the experiment in 30 seconds

MAKE THE VERIFICATION TARGET EXPLICIT

INPUT

Random information BITS

or an image. CV only decides which 8×8 blocks use 4 or 8 bits.

LINK

QPSK modulation

AWGN, Rayleigh and Rician channels. The RX estimates h, equalizes, hard-demaps and runs Viterbi when coded.

EVIDENCE

BER curves with theory

95% confidence intervals, coding gain, constellation, spectrum and image-layer PSNR.

One sentence: validate the link with theory first, then show coding and CV payload trade-offs.

What is inside the link?

EVERY BOX MAPS TO A CODE BLOCK

TX

bits → parity → QPSK mapper

CHANNEL

h·s + n
AWGN / Rayleigh / Rician

RX

pilot estimate → equalizer → hard decision

FEC

Viterbi → information bits

METRICS

BER · CI95
PSNR · rate

APPLICATION ENTRY

Fixed 6-bit image source ↔ CV adaptive 4/8-bit source

Why use Eb/N0 on the x-axis?

PUT MODULATION AND CODING ON THE SAME SCALE

QPSK: 2 bits / symbol

$E_s = 1$

$N0 = R / (2 \cdot 10^{(Eb/N0 / 10)})$

$R = 1$ uncoded

$R = 1/2$ convolutional code

PROBLEM

Using Es/N0 mixes the 2 bits/symbol factor into the comparison.

FIX

Use information-bit Eb/N0. AWGN BPSK and Gray-QPSK then share the same per-bit theory curve.

DEFENSE LINE

Normalize information-bit energy first, then compare channel and coding effects.

What changes in a Rician channel?

K IS THE LOS-TO-SCATTER POWER RATIO

```
K = 10^(K_dB / 10)
h = √(K/(K+1))
    + √(1/(K+1)) · g
g ~ CN(0,1)
```

K = 0

No LOS term. This becomes Rayleigh.

K = 10 dB

LOS power is ten times diffuse power.

K → ∞

Fading vanishes as the channel approaches AWGN.

The previous zero-mean complex Gaussian model was Rayleigh. The new code fixes this.

How is Monte Carlo made credible?

SAMPLE COUNT, FIXED SEED AND CONFIDENCE INTERVALS

SWEEP

Eb/N0 = 0:2:16 dB

AWGN / Rayleigh / Rician

uncoded / coded

SAMPLES

28 independent trials

12,000 information bits / trial

336,000 information bits / point

UNCERTAINTY

$\hat{p}$ = errors / bits

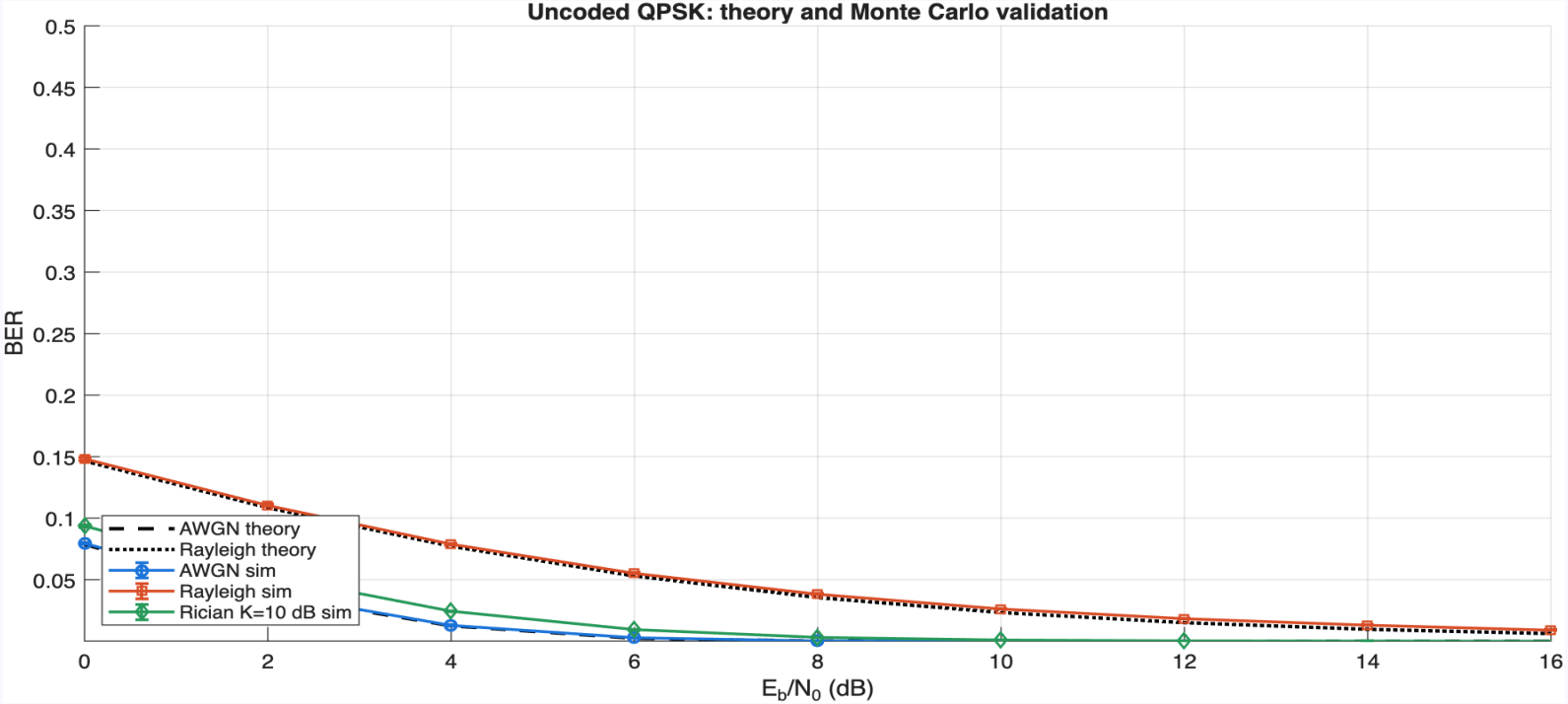
$CI95 = 1.96 \times \sqrt{[\hat{p}(1-\hat{p})/N]}$

fixed rng seed

At high SNR, report the confidence interval when zero errors are observed.

Core result 1: theory and simulation agree

THIS VALIDATES THE SIMULATOR BEFORE DISCUSSING CODING GAIN



AWGN

Markers follow Q(sqrt(2γ)).

Rayleigh

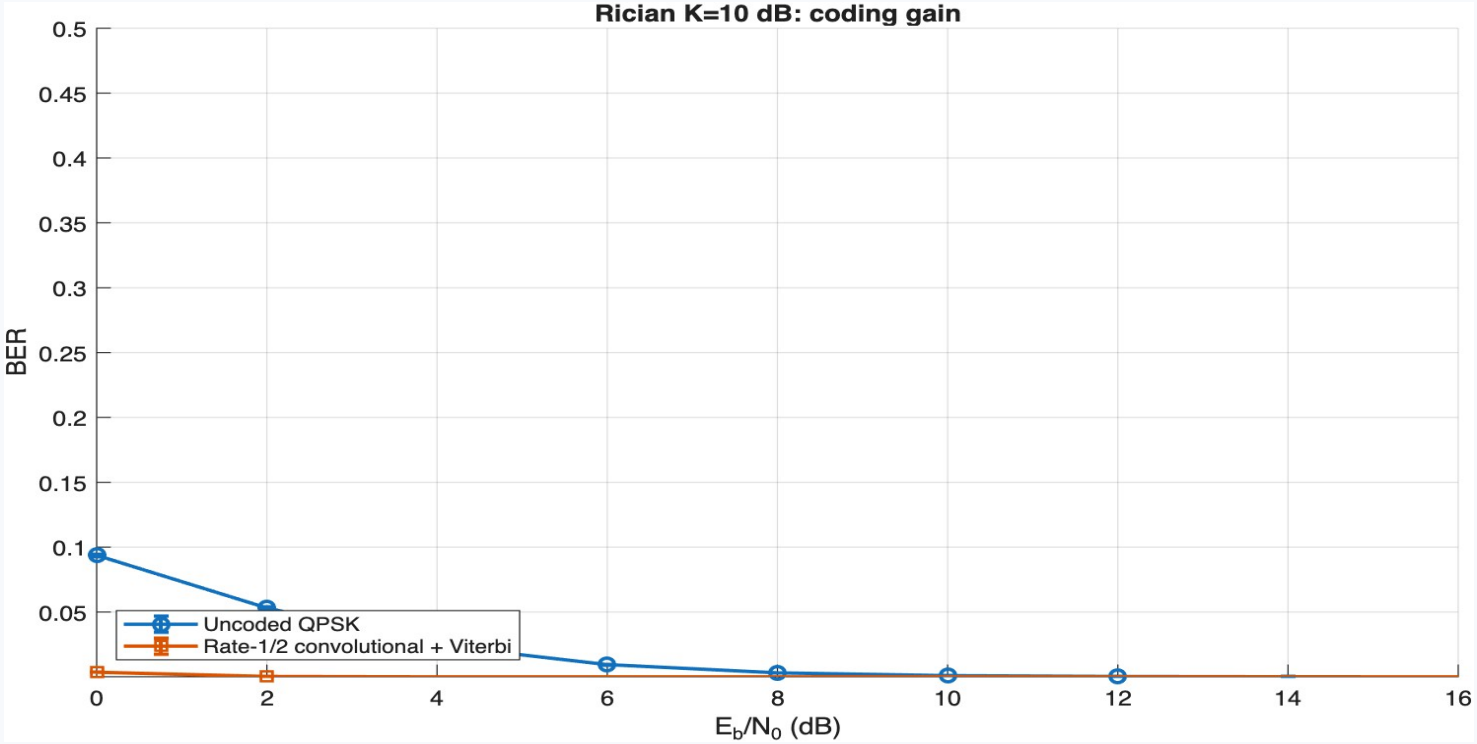
Markers follow the Rayleigh theory expression.

Rician

K=10 dB adds LOS power, so BER is lower than Rayleigh.

Core result 2: how much gain does coding buy?

RATE-1/2 CONVOLUTIONAL CODE + VITERBI



$g1 = 111_2$
$g2 = 101_2$

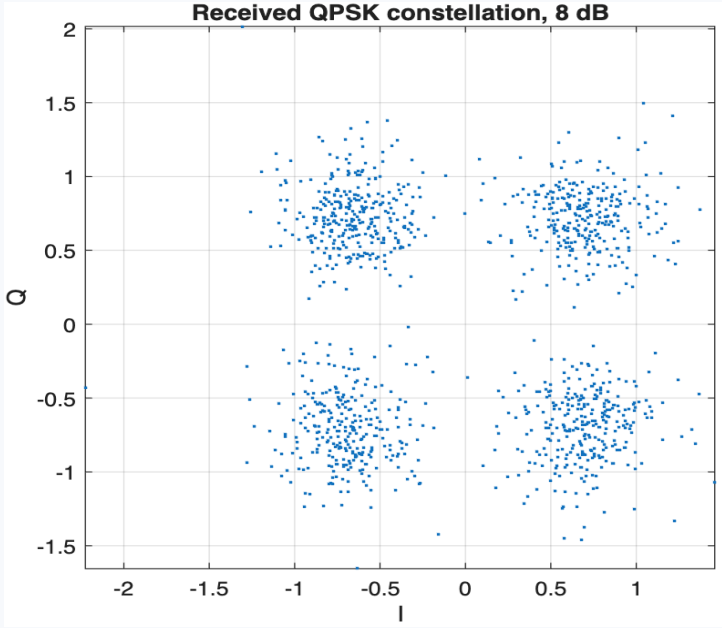
4 states · hard Viterbi

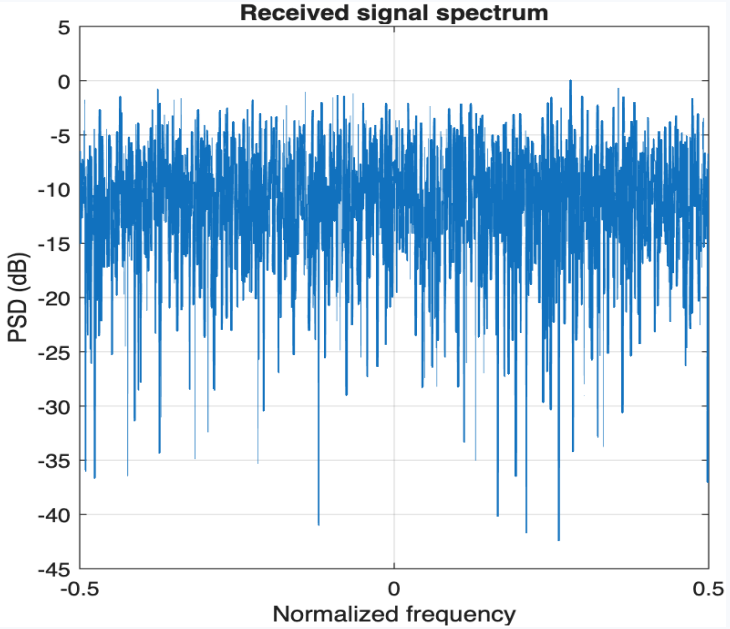
RESULT

At Rician 8 dB, uncoded BER≈3.1×10^{-3}. The coded run has zero observed errors in 336,000 information bits.

Waveform evidence: the link really carries QPSK

CONSTELLATION AND SPECTRUM MAKE THE SIGNALING WAVEFORM VISIBLE





CONSTELLATION

Rician fading spreads the four ideal points before equalization.

SPECTRUM

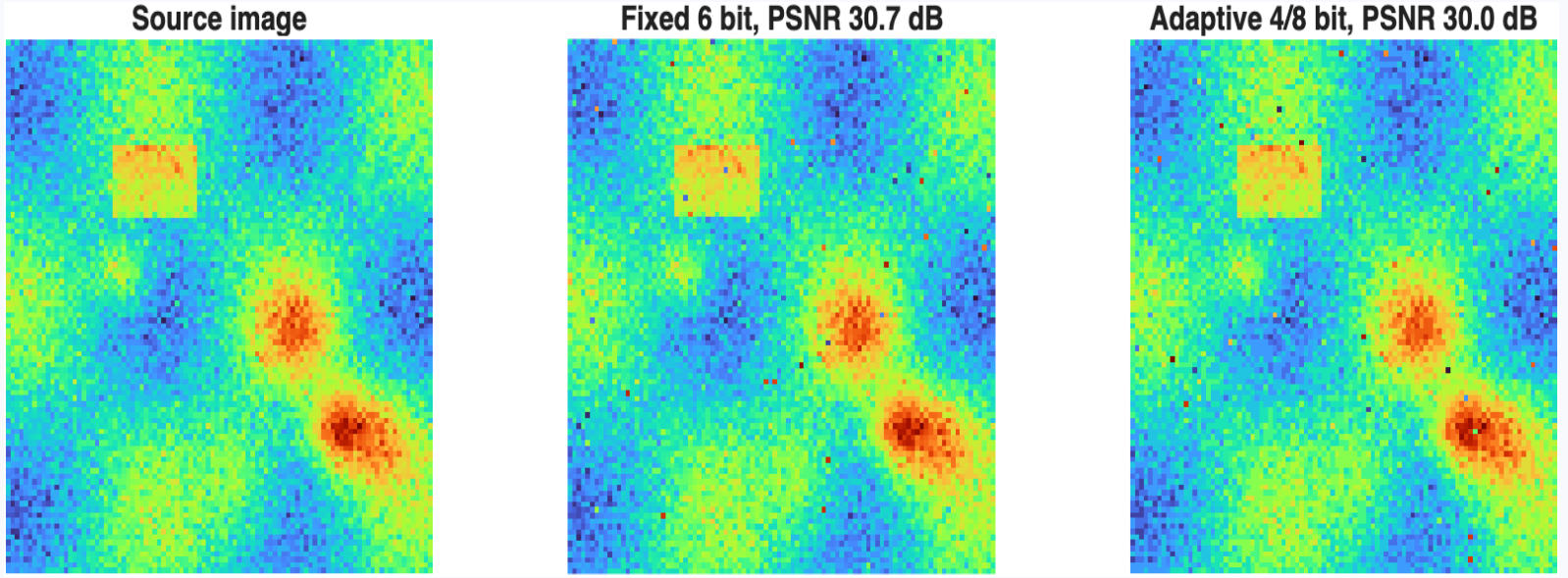
The FFT shows signal energy over normalized frequency.

RECEIVER

A small h-estimation error models imperfect pilots.

Where does CV fit?

IT IS A PAYLOAD POLICY INSIDE THE TRUSTED LINK



FIXED PAYLOAD

6 bits per pixel.

ADAPTIVE PAYLOAD

Gradient complexity selects 4 or 8 bits per 8×8 block.

SAME LINK

Both payloads use the same QPSK, Rician channel and Eb/N0.

Fixed PSNR≈30.7 dB. Adaptive PSNR≈30.0 dB, with target-region PSNR≈32.3 dB.

How do you read the code?

TRACE FROM THE ENTRY POINT TO BER

run_project_v3.m

1 link budget
2 BER Monte Carlo loop
3 qpskChannel()
4 convEncode()
5 viterbiDecode()
6 waveform figures
7 transmitImage()
8 write CSV + MAT

START HERE

qpskChannel() maps QPSK, adds noise and h, estimates h_hat, equalizes and hard-demaps.

NEXT

viterbiDecode() is a four-state dynamic program with path metrics and traceback.

OUTPUT FILES

ber_validation.csv feeds the theory plot. link_budget.csv stores the link budget. application_layer.csv stores the CV payload results.

Remember three sentences

CLOSE THE PROJECT WITH ONE STORY

1

I validate the QPSK simulator with AWGN and Rayleigh theory before discussing Rician.

2

I measure coding gain with a rate-1/2 convolutional code and Viterbi.

3

I connect a CV-adaptive image source to the same trusted link.

QPSK is the main character. Rician fading and coding provide depth. CV is the application layer.

Q&A